

Nintendo DS – R4 / R4i / DSTT / Ace3DS Guide

Recommended links

-  Recommended creator – NanospeedGamer:
[Visual guides for R4 cards – NanospeedGamer](#)
Clear and structured videos to follow the process visually.
-  Official kernel sites (examples):
Each flashcart model has its own official website or repository to download the internal system (kernel). Always use the official source.
-  Recommended accessories (Amazon + AliExpress):
 -  Recommended R4 card:
[Amazon \(R4 search\)](#) | [AliExpress \(R4 search\)](#)
 -  Recommended microSD card:
[Amazon \(microSD 32GB\)](#) | [AliExpress \(microSD 32GB\)](#)

Recommended material (affiliate links)

These links may contain affiliate tags. Replace only the affiliate ID to adapt them to your account.

- R4i Gold 3DS RTS (recommended):
[Buy R4i Gold 3DS RTS on Amazon](#)
- Ace3DS Plus:
[Buy Ace3DS Plus on Amazon](#)
- DSTT (TTDS):
[Buy DSTT on AliExpress](#)
- microSD 16GB / 32GB:
[Buy recommended microSD on Amazon](#)
- USB microSD reader:
[Buy USB reader on AliExpress](#)

Important notice

- Guide intended for personal and educational use.
- Use only personal backups of games you legally own.
- No ROMs, BIOS files, or copyrighted content are distributed or linked.
- The console is not modified; the flashcart handles everything.
- Back up important data before formatting your microSD.

What is an R4 card and what does it do?

An **R4 card** (or variants like R4i, DSTT, Ace3DS) is a special cartridge inserted into the Nintendo DS game slot, just like an original game.

It allows you to:

- Store multiple personal backups on a single microSD card.
- Run homebrew apps and games.
- Reduce wear on original cartridges.
- Use inexpensive microSD cards to organize your library.

The console is never modified; the flashcart provides all special functionality.

Compatibility

Compatible console models

Console	Compatibility	Notes
Nintendo DS	✓ High	Most original R4 cards work without issues.
Nintendo DS Lite	✓ High	Very common model for flashcart use.
Nintendo DSi	✓ Partial	Requires specific R4i models compatible with DSi.
Nintendo 3DS / 2DS	✓ Partial	Only some modern R4i cards work; this guide focuses on DS/DS Lite.

For simple, trouble-free use, DS or DS Lite is recommended.

Choosing a card

Which card should you choose?

General recommendation:

- **R4i Gold 3DS RTS (Wood R4)** – stable model with good support.

Other common cards:

- **Ace3DS Plus** – simple and compatible with many models.
- **DSTT (TTDS)** – widely used, requires its specific kernel.
- **R4 SDHC Dual-Core** – functional but with many kernel variations.

Avoid:

- Cards with generic labels like “R4 2024”, “R4 Upgrade”, “5000 in 1”.
- Models without a clear official website.

Required material

Before you begin

- A compatible **R4 / R4i / DSTT / Ace3DS** card.
- A **microSD** card (8–32 GB recommended).
- A **Windows or Mac PC**.
- A **microSD reader** (USB or built-in).
- A **Nintendo DS or DS Lite** with enough battery.

Step 1 – Prepare the microSD card

- 1 Connect the microSD card to your PC using a reader.
- 2 If it contains important data, make a backup.
- 3 Format the microSD as **FAT32**:
 - Windows: right-click → “Format...” → File system: FAT32.
 - If FAT32 does not appear, use a tool like “GUIFormat”.
- 4 Download the **kernel** for your specific flashcart from the official website.
- 5 Extract the kernel files and copy them to the root of the microSD (not inside a folder).
- 6 Create a folder named `games` in the root of the microSD.
- 7 Safely eject the microSD.

Example microSD contents

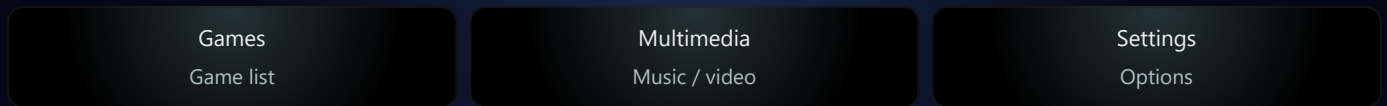
```
/R4iMenu/  
/TTMenu/  
/_rpg/  
/games/  
WOODR4.NDS  
TTMENU.DAT
```

Exact structure varies by card, but must include the kernel files and the `games` folder.

Step 2 – First boot on the Nintendo DS

- 1 Insert the microSD into the R4 card.
- 2 Insert the R4 card into the DS game slot.
- 3 Turn on the console and select the cartridge icon.

Flashcart main menu (example)



Menu design varies, but structure is similar.

If the menu loads correctly, the microSD and kernel were prepared properly.

Step 3 – Add your backups

- 1 Connect the microSD back to the PC.
- 2 Open the games folder.
- 3 Copy your personal backups and/or homebrew into this folder.
- 4 Safely eject the microSD and insert it into the R4.
- 5 Turn on the DS and open the Games menu.

Example game list

- New Super Mario Bros.nds
- Mario Kart DS.nds
- The Legend of Zelda – Phantom Hourglass.nds
- Pokémon White.nds

Select a title with the D-pad and press A to launch.

Common errors and fixes

“MENU?” message

“MENU?”

The flashcart cannot find the internal system files (kernel) required to boot.

- Verify you downloaded the correct kernel for your exact flashcart model.
- Ensure the kernel files are placed in the **root** of the microSD.
- If needed, reformat the microSD to FAT32 and copy the kernel again.

“FAT Error” message

“FAT Error”

Indicates a problem with the microSD file system.

- Back up any recoverable data.

- Format the microSD in FAT32.
- Copy the kernel and the `games` folder again.

“Unsupported DLDI” or game not launching

Compatibility error

The game or homebrew is not compatible with your flashcart or current kernel.

- Check if a newer kernel version exists.
- Look for a compatible version of the homebrew/game for your card.

Flashcart does not boot on DSi or 3DS

DSi/3DS incompatibility

Some R4 models are not designed to work on DSi or 3DS systems.

- Use a card specifically marked as compatible with DSi/3DS (e.g., R4i Gold 3DS RTS).
- Alternatively, use the flashcart on a Nintendo DS or DS Lite.

Optimization and savings

Practical tips

- A single flashcart can serve multiple users on the same console.
- Store original cartridges safely and use the flashcart for daily play.
- 16GB or 32GB microSD cards are inexpensive and hold many titles.
- Legal homebrew expands console functionality at no extra cost.